# 01 - Count and accumulate

## Read, predict and model

Name _____ Class _____ Date _____

Trace a counted loop and distinguish a counter from a running total.

Textbook: loops

Before starting: Complete the foundations pack: assignment, types, comparisons, selection and boundary tests. Teach these four activities in order; revisit tracing before adding repetition.

A club records points of 2, 4 and 6. range(1, 4) supplies 1, 2, 3; the stop value is excluded. Start total outside the loop so earlier values are preserved.

```
total = 0
for round_number in range(1, 4):
    total = total + round_number * 2
print(total)
```

| Round | Added | Total after update |
|---|---|---|
| 1 | 2 | 2 |
| 2 | 4 | 6 |
| 3 | 6 | 12 |

The loop runs three times and prints 12. Moving total = 0 inside the loop would discard earlier additions.

## Equipment-free route

Use a pencil and a trace table, treating each loop iteration as one row. Record hand-traced results explicitly; do not label them as executed Python output.

## Explain and transfer

Annotate one step of the worked example in your own words. Change one supplied input, record a new prediction and show how the rule produces it. Label a hand trace separately from a software observation.

# 01 - Count and accumulate

## Independent practice and evidence

Name _____ Class _____ Date _____

### A. Predict and trace

Change the stop value to 5. Draw a trace with round number, added points and total. Predict the final output before running or hand-executing.

### B. Independent program

Write a loop to add the integers from 1 through n inclusive. Use supplied n = 4, then test n = 0 and n = 1. Record expected and actual totals separately. Explain why the accumulator starts at zero.

| n | Expected | Actual / hand trace |
|---|---|---|
| 4 | | |
| 0 | | |
| 1 | | |

### Exit

A learner uses range(1, n). Give a test that exposes the missing final value and explain the fix.

# 02 - Return a result from a function

## Read, predict and model

Name _____ Class _____ Date _____

Separate parameters, return values and printed output.

Textbook: functions

This function applies a fixed booking charge of $2 and $3 per item. Assume quantity is a non-negative integer. The parameter is local to the function; return sends a value back to its caller.

```
def booking_cost(quantity):
    return quantity * 3 + 2

first = booking_cost(4)
print(first)
```

quantity receives 4, the expression evaluates to 14, first receives 14 and print displays it. Defining the function alone does not perform this call.

## Equipment-free route

Act out the function with an input card and a separate return card. Calculate all three calls by hand and label the results as traced.

## Explain and transfer

Annotate one step of the worked example in your own words. Change one supplied input, record a new prediction and show how the rule produces it. Label a hand trace separately from a software observation.

# 02 - Return a result from a function

## Independent practice and evidence

Name _____ Class _____ Date _____

### A. Trace two calls

Predict booking_cost(0) and booking_cost(5). Explain why replacing return with print would stop first receiving the calculated number.

### B. Independent transfer

Write cover_cost(quantity, price) returning quantity * price + 4. Test (0, 2), (3, 2) and (2, 5). Keep the function separate from printing. State the allowed input domain.

| Arguments | Expected return | Observed / traced return |
|---|---|---|
| 0, 2 | | |
| 3, 2 | | |
| 2, 5 | | |

### Exit

Explain the difference between a parameter and an argument using one of your calls.

# 03 - Traverse a list and retain a position

## Read, predict and model

Name _____ Class _____ Date _____

Use zero-based indices and handle empty input deliberately.

Textbook: arrays

The task is to find the index of the first smallest value in a non-empty list. Start with the first valid index. Use <, rather than <=, to preserve the first equal minimum.

```
values = [8, 3, 3, 6]
best = 0
for index in range(1, len(values)):
    if values[index] < values[best]:
        best = index
print(best, values[best])
```

At index 1 the minimum becomes 3; the equal 3 at index 2 does not replace it. The output is 1 3. The initial index is invalid for an empty list.

## Equipment-free route

Cut out value cards and number their positions from zero. Move a best-index marker only when the comparison is true.

## Explain and transfer

Annotate one step of the worked example in your own words. Change one supplied input, record a new prediction and show how the rule produces it. Label a hand trace separately from a software observation.

# 03 - Traverse a list and retain a position

## Independent practice and evidence

Name _____ Class _____ Date _____

### A. Independent trace

Trace [5, 7, 2, 2]. Show index, comparison and best after every iteration. Explain how <= would change the answer.

### B. Extend the contract

Add selection before the algorithm: if the list is empty, print 'No data'; otherwise find the first minimum. Test [], [9] and [4, 1, 1]. Do not silently report index 0 for an empty list.

| Data | Expected | Actual / traced |
|---|---|---|
| [] | | |
| [9] | | |
| [4, 1, 1] | | |

### Exit

Distinguish index 1 from the value stored at index 1.

# 04 - Read a file and test bad records

## Read, predict and model

Name _____ Class _____ Date _____

Read sequential text, convert values and report unsuitable records.

Textbook: files

Assume a UTF-8 text file called readings.txt exists in the working folder. It contains one integer per line. The example ignores blank lines; a non-integer line is reported and skipped. File-not-found recovery is a separate task.

```
total = 0
with open("readings.txt", encoding="utf-8") as source:
    for line in source:
        text = line.strip()
        if text != "":
            try:
                total += int(text)
            except ValueError:
                print("Invalid:", text)
print("Total:", total)
```

For lines 3, blank, x, 5 the output is Invalid: x followed by Total: 8. int converts suitable text; the exception branch does not add a value.

## Equipment-free route

Use the printed record lines as the file. Trace the accumulator and output queue line by line; no editor or device is required.

## Explain and transfer

Annotate one step of the worked example in your own words. Change one supplied input, record a new prediction and show how the rule produces it. Label a hand trace separately from a software observation.

# 04 - Read a file and test bad records

## Independent practice and evidence

Name _____ Class _____ Date _____

### A. Independent trace

Trace lines 2, -1, bad, 4. List the output messages in order. The current program accepts negative integers; distinguish conversion from a domain rule.

### B. Change the requirement

Only readings 0 through 10 inclusive may contribute. Add a range test after conversion. Print Out of range: followed by the integer when outside this range. Preserve invalid-text reporting. Test lines 0, 10, 11, -1, oops, blank.

| Case | Expected message / total effect | Actual / traced |
|---|---|---|
| 0 and 10 | | |
| 11 and -1 | | |
| oops and blank | | |

### Exit

Does this program recover from a missing file? Explain where that failure occurs.